CINEAI LAB

Experimental Foundations for Open AI-Native Cinema

From Generated Clips to Image-Directed Filmmaking

DIRECTOR, CineLingo and an image-first architecture for controllable AI cinema

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English reference edition

Creative and research direction: Rene Boroughs

Editorial, research and technical collaboration: OpenAI ChatGPT

https: cineai!lab"org

Independent project. No affiliation with the tool or model providers discussed in this manuscript.

Status and reading note

This is a #or$ing research %anuscript, not a product specification, a technical bench%ar$ or a clai% that a co%plete AI! native feature!fil% pipeline has been solved" CineAI &ab uses narrative pro'ects as practical stress tests" The purpose is to docu%ent #hat current tools %a$e possible, #here the(fail under fil%%a$ing constraints, and #hat $ind of authoring environ%ent appears necessar(#hen a creator #ants continuit(, selective revision and durable pro'ect %e%or("

> **The central change in version 0.10**
>
> The pro'ect has %oved fro% as$ing ho# to generate convincing clips to#ard as$ing ho# to preserve a fil%%a$er)s decisions #hile AI creates onl(the %ove%ent or transfor%ation that has been authorised" This shift leads to an i%age!first %odel: persistent source i%ages, shot instances, na%ed selections, te%poral behaviours and replaceable AI engines"

What changed since version 0.9

- The e*peri%ental record no# includes local &T+!,"- #or$ and later Co%f(. I / an ,", i%age!to!video tests"
- The #or$ing %ethod has shifted to#ard validated still i%ages as persistent pro'ect assets rather than disposable pro%pts"
- 0IRECTOR is no# described as a database!bac$ed authoring interface under develop%ent, not %erel(as an abstract open!fil% aspiration"
- Cine&ingo is introduced as a natural!language orchestration la(er inspired b(the separation of Cast, 1tage, 1core and Behavior in 0irector &ingo"
- The 2issing Piece has beco%e the principal i%age!first fil% de%onstrator"
- The technolog(#atch no# focuses on first last!fra%e control, interpolation, faster iteration, provenance and the integration of AI into creative #or$spaces"

Authorship and AI assistance

The protected narrative #or$s, their characters, dra%atic choices and final creative decisions re%ain the responsibilit(of their hu%an author" ChatGPT has been used as a sustained research and develop%ent collaborator for screenpla(anal(sis, e*peri%ental design, docu%entation, tool co%parison, debugging, architecture #or$ and editorial consolidation" This assistance does not transfer authorship of the narrative #or$s and does not i%pl(affiliation bet#een CineAI &ab and OpenAI"

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Executive summary

Artificial intelligence can no# generate stri$ing still i%ages, s(nthetic voices and short %oving se<uences" =et the production of an editable, coherent fil% re%ains a different proble%" A fil%%a$er does not onl(need a clip that loo$s plausible" The fil%%a$er needs to decide #hat the shot is, preserve characters and ob'ects across ti%e, direct perfor%ance, return to an earlier shot #ithout losing continuit(, and change one ele%ent #ithout accidentall(changing ever(thing else"

CineAI &ab began b(e*peri%enting #ith the available co%ponents of this e%erging ecos(ste%: still!i%age develop%ent, ,"-0 ca%era %otion, P(thon and 99%peg pipelines, s(nthetic voice, %ultilingual vise%es, facial! ani%ation routes, local video generation #ith &T+, and later Co%f(. I / an i%age!to!video #or$flo#s" 1everal e*peri%ents produced valid files" 9e#er produced convincing cine%a" This distinction bet#een technical success and cine%atic success beca%e the research ob'ect"

The %ost i%portant conclusion is architectural" In %an(current generative #or$flo#s, the %odel is treated as the place #here the scene is created" The user provides an i%age or pro%pt and the %odel invents the ne*t fra%es" 9or a fil%%a$er #ho has alread(designed the i%age, this can be bac$#ards" If the face, hand position, co%position, lighting, bac$ground and ca%era have alread(been chosen, regenerating the entire fra%e to create a s%all %ove%ent is both visuall(ris$(and co%putationall(e*pensive"

CineAI &ab therefore proposes an i%age!first #or$flo#" A fil% can be understood as a se<uence of deliberatel(chosen still states, %uch li$e a highl(sophisticated flipboo$" The fil%%a$er %a($no# the state at the beginning of a shot and the state at the end" The useful role of AI is then to create the %issing %ove%ent bet#een those states #hile respecting #hat has not changed"

> **Working proposition**
>
> I%age A > selected %ove%ent > I%age B" The fil%%a$er defines the i%ages, the ele%ents allo#ed to %ove, the ele%ents that %ust re%ain loc$ed and the intended ti%ing" The AI engine e*ecutes the transition inside those li%its"

This proposition led to the develop%ent of 0IRECTOR, an interface concept that stores i%ages as persistent assets, places the% as instances in shots, attaches %as$s and loc$s to selected regions, represents %ove%ent as ti%eline events, and delegates e*ecution to replaceable engines" Cine&ingo co%ple%ents 0IRECTOR as a hu%an!readable orchestration la(er" Rather than as$ing a creator to %anipulate a node graph, it translates directions such as ?at 3, seconds, %ove the right hand slo#l(@ $eep the ca%era fi*edA into structured te%poral actions and constraints"

The approach deliberatel(separates fil%, interface and engine" The fil% is the durable creative ob'ect: scenes, i%ages, dialogue, ti%ing, continuit(, selections and behaviours" 0IRECTOR is the authoring interface" &T+, Co%f(. I, / an, 2useTal$, Blender, 99%peg or future %odels are e*ecution engines and adapters" If one engine disappears or i%proves, the fil% should survive"

The 2issing Piece is no# the %ain practical de%onstrator for this approach" It contains recurring characters across age and geograph(, fa%il(group scenes, inti%ate %icro!perfor%ance, technical environ%ents and a recurring puBBle %otif" These ele%ents %a$e it useful for testing e*actl(the proble%s that generative video still struggles #ith: identit(preservation, ense%ble continuit(, selective %otion and the abilit(to revise a shot #ithout redra#ing the #orld"

Cersion ;"3; therefore refra%es CineAI &ab" The pro'ect is no longer pri%aril(searching for a single %odel capable of ?%a$ing the fil%"A It is investigating a fil%%a$er!controlled environ%ent in #hich %ultiple AI s(ste%s can be used as instru%ents #hile the pro'ect %e%or(, visual decisions and dra%atic structure re%ain under hu%an control"

1. Cinema needs more than generated clips

1.1 A clip is an output; a film is a revisable system

Generative video de%onstrations are often evaluated one clip at a ti%e" Cine%a, ho#ever, is cu%ulative" The credibilit(of shot ,7 depends on #hat the audience sa# in shots ,, : and 38" A costu%e detail, a piece of furniture, the age of a character, the direction of a gaBe or the position of an ob'ect %a(carr(narrative %eaning" These ele%ents %ust persist even #hen the tool used to render the shot changes"

&ong!for% #or$ also de%ands revision" A director %a(#ant to change onl(the hesitation before a line, e*tend a hand gesture b(half a second, hold the ca%era still, or replace one bac$ground ob'ect" A s(ste% that responds b(regenerating the entire fra%e behaves less li$e an editing environ%ent and %ore li$e a lotter(#hose tic$et is a pro%pt"

1.2 The filmmaker already knows more than the model

In %an(practical situations the fil%%a$er is not starting fro% nothing" The character has alread(been designed" The co%position has been approved" The lighting, lens feeling, decor and e%otional beat have been chosen" The challenge is not invention but controlled transfor%ation" This is #h(i%age!to!video is interesting but insufficient #hen the onl(strong anchor is the first fra%e"

The $e(fil%%a$ing <uestion is therefore not %erel(?Can the %odel generate %otionDA but ?Can the s(ste% preserve all decisions that are outside the re<uested %otionDA

1.3 Human direction as explicit data

Eu%an control is often described in broad ethical ter%s" CineAI &ab treats it as a concrete data proble%" 0irection should be represented e*plicitl(: #hich character is present, #hich i%age is the approved source, #hich region %a(%ove, #hich region is loc$ed, #hat happens at #hich ti%e, #hat continuit(rule %ust hold, and #hich $inds of failure should invalidate a render" If those decisions e*ist onl(inside a pro%pt, the(are difficult to audit, revise or %ove bet#een engines"

> **Design principle**
>
> The pro'ect should re%e%ber the director)s decisions independentl(of the %odel that happened to render a particular version"

2. Experimental method: film problems first

2.1 Narrative projects as stress tests

CineAI &ab does not begin #ith a leaderboard of %odels" It begins #ith scenes" A scene i%poses re<uire%ents that are difficult to hide: the sa%e face %ust return, a hand %ust touch a specific ob'ect, a pause %ust last long enough, an ani%al %ust appear to have #eight, or a group %ust react #ithout the bac$ground %elting" The narrative pro'ect supplies the pressure@ the e*peri%ent reveals the technical gap"

2.2 The difference between file success and cinematic success

A render can co%plete #ithout errors and still fail as cine%a" Conversel(, a %odest technical result can be useful if it reveals a %issing control" CineAI &ab therefore records both la(ers: #hether the pipeline ran, and #hether the result respected the intended shot"

Period	Experiment	Question	Main observation
FunGFul ,;,6	1till i%ages and visual anchors	Can character and co%position be fi*ed before %otionD	=es" Approved stills offered the strongest continuit(and beca%e reliable shot anchors"

FulGAug , ; , 6	Blender P(thon 99%peg ,"- 0	Can controlled ca%era %ove%ent create cine%atic lifeD	Ca%era and paralla* #ere controllable, but %otion of the fra%e did not e<ual actor perfor%ance"
FulGAug , ; , 6	Coice, vise%es, lip!s(nc	Can speech beco%e directed perfor%anceD	Ti%ing and %outh!shape data #ere feasible@ gaBe, breathing, anato%(and e%otional rh(th% re%ained harder"
Aug , ; , 6	&T+!,"- local inference	Can an open video %odel run locall(on Apple 1iliconD	&ocal inference #as technicall(possible #ith aggressive offloading, but production control and speed re%ained li%iting"
1epGOct , ; , 6	Co%f(. I / an ,", I, C	Can a validated i%age beco%e a stable %oving shotD	The tested #or$flo# used a starting i%age plus pro%pt@ #hole!shot reinterpretation and long iteration ti%es under%ined precise direction"
Oct , ; , 6	0IRECTOR Cine&ingo	/ hat should the authoring la(er re%e%berD	I%ages, instances, selections, loc$s, ti%ing and behaviours should persist independentl(of the rendering engine"

2.3 Hardware realism matters

E*peri%ents #ere deliberatel(perfor%ed on consu%er!class hard#are as #ell as through hosted tools" This is i%portant because a fil%%a$er!oriented #or$flo# cannot be 'udged onl(on an ideal high!%e%or(GP . " / aiting %an(%inutes for a fe# seconds of uncertain output changes the creative process itself" 0irection depends on iteration, and iteration depends on feedbac$ speed"

3. What the experiments taught us

3.1 Still images: the most stable source of authorship

The still i%age repeatedl(proved to be the point at #hich the fil%%a$er had the clearest control over co%position, character identit(and production design" Once a still #as approved, treating it as disposable %ade little sense" The natural conclusion #as to elevate approved i%ages fro% te%porar(pro%pts to persistent pro'ect assets"

3.2 2.5D motion: precise but not performance

&ocal ,"- 0 tests confir%ed that ca%era %otion, paralla*, co%positing and e*port can be %ade deter%inistic" This is valuable for transitions, at%ospheric %ove%ent and certain docu%entar(or graphic st(les" But it also e*posed a conceptual li%it: %oving a ca%era over a still i%age does not %a$e a character act" The bod(needs internal %ove%ent, #eight transfer, gaBe and ti%ing"

3.3 Voice and facial animation: timing is easier than acting

1(nthetic speech, vise%e se<uences and lip!s(nc pipelines can be structured and ti%ed" The difficult part is not the e*istence of %outh %ove%ent but the coherence of perfor%ance" 1peech includes breath, hesitation, listening, gaBe, posture and e%otional intention" These di%ensions need to be directed together rather than generated as unrelated effects"

3.4 LTX-2.5: local inference was possible, but not yet a complete production path

The August , ; , 6 &T+!,"- e*peri%ent de%onstrated that local inference on a ,5 GB Apple 2 4 could co%plete through a native P(thon pipeline #ith aggressive offloading, even though the co%bined %odel co%ponents e*ceeded available unified %e%or(" This #as an i%portant technical result" It did not, ho#ever, solve the fil%%a$ing proble%: generation re%ained slo#er than interactive direction and the resulting %otion did not consistentl(preserve the desired i%age #ith sufficient precision for the current fil%"

> **Interpretation**
>
> A successful inference proves that a %odel can run" It does not prove that it belongs at the centre of a fil%%a$ing interface"

3.5 ComfyUI / Wan 2.2: the cost of regenerating too much

&ater tests %oved to Co%f(. I and / an ," , i%age!to!video #or$flo#s" In the tested configuration, the #or$flo# accepted a starting i%age and a te*tual description of #hat should happen ne*t, but it did not provide a true fil%%a$er! defined final i%age as a second hard anchor" The %odel therefore had to invent the continuation"

9or the $inds of shots being developed in The 2issing Piece, this #as often the #rong trade!off" A desired %icro!action could be acco%panied b(face drift, altered hands, changed ob'ects, ne# ca%era %otion or a reinterpretation of the bac$ground" The proble% #as a%plified b(generation ti%e: a fe# seconds of footage could re<uire enough setup and rendering ti%e that repeated s%all corrections beca%e i%practical"

The conclusion is deliberatel(li%ited to the #or$flo#s actuall(tested" It is not a clai% that Co%f(. I or / an cannot support other for%s of control" It is a record of #h(that path #as not accepted as the central CineAI &ab production %ethod at this stage"

3.6 The useful failure

These e*peri%ents did not %erel(eli%inate tools" The(clarified the %issing la(er" The core re<uire%ent is not another generator" It is an authoring environ%ent capable of telling an(generator #hat it is allo#ed to change, #hat it %ust preserve, and #here the shot %ust arrive"

4. The image-first turn: film as a flipbook

4.1 The simplest mental model

At a funda%ental level, %oving i%ages are se<uences of still fra%es" The analog(is intentionall(si%ple: a fil% is a ver(sophisticated flipboo$" Traditional cine%a captures or constructs those fra%es through photograph(, ani%ation, visual effects and editing" Generative AI adds a ne# #a(to create inter%ediate fra%es, but it does not re%ove the need for endpoints, continuit(and directorial choice"

This %odel changes the role of AI" Instead of treating the first i%age as a suggestion and the re%ainder of the shot as free space, both i%ages can be treated as deliberate states" The engine)s 'ob beco%es to construct a plausible, te%porall(coherent path bet#een the%"

4.2 Movement should be local when the intention is local

A father %a(point at a %issing puBBle piece #hile the table, bac$ground and other fa%il(%e%bers re%ain largel(unchanged" A child %a(shift his gaBe #hile the ca%era sta(s fi*ed" A hand %a(return to a lectern after a sentence" These are local events" Reco%puting ever(pi*el as if the entire shot #ere uncertain is #asteful and invites drift"

4.3 Start image, end image and intermediate keyframes

The preferred #or$flo# therefore includes at least a start i%age and an end i%age, #ith optional inter%ediate $e(fra%es #hen the action is co%ple*" The $e(fra%es are not si%pl(conditioning i%ages@ the(are directorial co%%it%ents" The s(ste% should treat the% as states to be respected"

4.4 Selected motion and locked regions

The fil%%a$er should be able to select a region or ob'ect and declare that it %a(%ove, #hile si%ultaneousl(loc$ing identit(, costu%e, fra%ing, ca%era and bac$ground" This transfor%s control fro% a vague pro%pt into an e*plicit contract"

> **Image-first proposition**
>
> The fil%%a$er asse%bles the i%portant states first" AI supplies continuit(bet#een the%, not per%ission to redesign the shot"

5. DIRECTOR: a filmmaker-facing authoring environment

5.1 Why the name matters

0IRECTOR is not used here as a generic s(non(% for ?control"A It refers to an interface architecture being developed #ithin CineAI &ab, inspired b(an earlier authoring insight found in 2acro%edia Adobe 0irector: separate the source %edia, the visible co%position, the te%poral score and the behaviours that act on instances"

5.2 From Director/Lingo to CineAI

Director / Lingo	CineAI DIRECTOR	Function
Cast	2edia Assets	Persistent source i%ages, video, audio, characters, sets and props
1tage	/ =1I / =G 1tage	E*act visible co%position of the shot
Cast %e%ber	1ource %edia	Original asset preserved independentl(of place%ent
1prite	1hot instance	Placed use of an asset #ith its o#n transfor%, ti%ing, %as$ and loc$s
1core	1e<uence Ti%eline	Te%poral organisation of events and instances
Behavior	Cine%atic behaviour	Reusable perfor%ance or %otion instruction
&ingo	Cine&ingo	Eu%an!readable orchestration translated to engine!specific actions
Pla(bac$ head	Pla(head	Current te%poral position inside the shot
2ovie	Pro'ect fil%	0urable creative ob'ect independent of rendering engine

5.3 Four visible zones

A practical 0IRECTOR interface can be organised around four visible Bones #ithout cop(ing 0irector)s historical appearance:

- 2edia Cast: the available i%ages, audio, characters, sets and props"
- 1tage: the e*act / =1I / =G co%position, #ith selectable and %ovable instances"
- Inspector: %as$s, feathering, loc$s, ani%atable regions and continuit(%etadata"
- 1core Ti%eline: dialogue cues, gestures, gaBe, e*pression, ca%era, audio, effects and loc$ events"

5.4 The Stage is the visual truth

The decisive rule is that the 1tage represents the fil%%a$er)s intended i%age" The AI %a(ani%ate selected properties, but it should not silentl(redefine the co%position" This borro#s the useful distinction fro% 0irector bet#een #hat the author la(s out visuall(and #hat a behaviour %a(%odif(at runti%e"

5.5 AI belongs behind the interface

The creative ob'ect should therefore re%ain readable even if a specific engine disappears" 0IRECTOR should not beco%e a si%plified cop(of Co%f(. I" Co%f(. I, #hen useful, should sit behind 0IRECTOR as one e*ecution route a%ong others"

6. A database model for film memory

6.1 Why a film needs structured memory

&ong!for% continuit(cannot depend on a folder of pro%pts" The pro'ect needs a durable representation of source %edia, their uses in individual shots, the events that occur over ti%e, the constraints applied to those events, and the renders produced b(different engines"

6.2 Proposed core entities

Entity	Purpose
assets	Persistent source %edia: i%ages, video, audio, %as$s, references
shots	The dra%atic and te%poral unit being directed
shotHinstances	A placed use of an asset inside a shot
dialogues	Te*t actuall(spo$en, language, spea$er and ti%ing
instructions	Positive directorial instructions
negativeHconstraints	Errors or transfor%ations that %ust be avoided
ti%elineHevents	Ti%ed behaviours such as gesture, gaBe, e*pression, ca%era or loc$
renders	Generated or co%posited outputs and their lineage
engineH'obs	E*ecution records for a specific rendering engine and configuration

6.3 Source asset versus shot instance

The distinction bet#een source and instance is essential" A source i%age should not be %utated si%pl(because it is placed in a shot" The instance o#ns its o#n position, siBe, rotation, opacit(, ti%ing, %as$, feathering, loc$s and ani%atable regions"

6.4 Timeline events as first-class data

T(pical event t(pes include gesture, gaBe, e*pression, head %otion, ca%era, audio, %usic, loc$, unloc$, %as$, behavior, dialogue cue, hold, transition and visual effect" This representation %a$es a director)s intentions <uer(able and portable"

6.5 Why a database is different from a node graph

A node graph describes ho# one e*ecution is asse%bled" A fil% database describes #hat the fil% is supposed to be" The t#o are related but should not be identical" I eeping the% separate prevents the pro'ect fro% beco%ing trapped inside a specific #or$flo# i%ple%entation"

> **Durability rule**
>
> A render can be disposable" A directorial decision should not be"

7. CineLingo: behaviours instead of prompt piles

7.1 A language for cinematic events

Cine&ingo is conceived as a fil%%a$ing orchestration la(er, not as a general!purpose progra%%ing language" The fil%%a$er describes cine%atic events in natural language" Cine&ingo resolves the affected instances, te%poral references, loc$s and ani%atable regions, then delegates e*ecution to the appropriate adapter"

7.2 Three levels of expression

The s(ste% distinguishes the hu%an direction, the structured Cine&ingo representation, and the engine!specific i%ple%entation" The fil%%a$er should nor%all(operate onl(at the first level"

Level	Example
0irector	?At the end, he loo$s at the ca%era #ith concern"A
Cine&ingo core	BE9ORE dialogue"end 3"-s > gaBe CA 2 ERA > e*pression concerned ;"4-
Engine adapter	2 odel!specific %as$s, pro%pts, controls, sa%plers, $e(fra%es or co%positing instructions

7.3 Locks and animatable regions

This is not cos%etic s(nta*" It e*presses the $e(contract bet#een director and %odel" A re<uest for a hand gesture should not i%pl(per%ission to change the face, costu%e or ca%era"

7.4 Temporal direction

7.5 Contradiction detection before generation

A structured language also per%its validation" If the ca%era is loc$ed and a later instruction re<uests a pan, the interface can flag the contradiction before e*pensive generation begins" The sa%e is true for a froBen hand follo#ed b(a gesture re<uest" Preventing i%possible or contradictor('obs is one #a(to reduce #asted co%putation"

7.6 Behaviours as reusable direction

Cine&ingo behaviours can encode recurring acting patterns #ithout freeBing the% into a single pro%pt" A ?restrained spea$erA behaviour %ight i%pl(subtle head %otion, lo# hand activit(, serious e*pression and audience!facing gaBe" Each engine %a(i%ple%ent those intentions differentl(, #hile the behaviour re%ains part of the fil% pro'ect"

8. Film, interface and engine must remain separate

8.1 Three layers

Layer	What it contains	What should survive
9il%	1cenes, shots, validated i%ages, dialogue, ti%ing, continuit(, selections, behaviours	Ever(thing creativel(%eaningful
Interface	0IRECTOR 1tage, Inspector, 1core, editing tools	The authoring logic and pro'ect readabilit(
Engine	&T+, / an, Co%f(. I #or$flo#s, 2 useTal$, Blender, 99%peg, future s(ste%s	Replaceable i%ple%entation details and render para%eters

8.2 Why engine independence matters

The AI video field changes <uic$l(" 2 odels appear, are replaced, change licensing, %ove bet#een hosted and local for%s, or disappear" A feature!length pro'ect %a(outlive several generations of tools" Treating the engine as a replaceable bac$end is therefore not onl(an open!source preference@ it is a basic production!resilience re<uire%ent"

8.3 Generation should be a job, not the project

An engine 'ob should record #hich shot version, assets, %as$s, constraints, behaviour events and engine para%eters #ere used to produce a render" The 'ob can fail, be retried or be superseded #ithout da%aging the creative source of truth"

8.4 Selective re-rendering

This architecture also %a$es selective revision possible" If a gesture changes, 0IRECTOR should be able to identif(the affected region and ti%e interval, create a ne# engine 'ob, and preserve the rest of the pro'ect" The long!ter% goal is not necessaril(pi*el!level partial diffusion in ever(engine@ it is a #or$flo# in #hich the scope of the re<uested change re%ains e*plicit"

9. The Missing Piece as the current demonstrator

9.1 Why this film is useful technically

The 2issing Piece is the current pri%ar(fil% pro'ect for testing i%age!first production" Its public description is intentionall(li%ited because the narrative #or$ re%ains protected" Technicall(, ho#ever, the pro'ect is de%anding in productive #a(s: the sa%e protagonist appears across ti%e and geograph(@ fa%il(and research!group scenes re<uire ense%ble continuit(@ a recurring puBBle i%age carries visual %eaning@ and later se<uences %ove into high!technolog(environ%ents #here s%all %echanical or biological changes %a(%atter"

9.2 The opening puzzle sequence

The opening se<uence provides a co%pact de%onstration of the 0IRECTOR proble%" A fa%il(is gathered around a puBBle" A father notices the %issing place" Attention shifts" The fa%il(leaves for a piJata" The child 2ateo later finds a piece and returns to the puBBle" 2uch of the shot design alread(e*ists as validated i%ages"

9or a transition bet#een t#o near!identical fa%il(i%ages, the desired change %a(be onl(a hand %ove%ent, a pointing gesture and subtle changes of gaBe and e*pression" The table, roo%, ca%era and %ost bodies should re%ain stable" This is precisel(the $ind of shot #here #hole!fra%e regeneration is an inefficient abstraction"

> **Demonstrator question**
>
> Can a s(ste% create the %issing fra%es bet#een t#o deliberatel(designed i%ages #hile changing onl(the selected actions and preserving the rest of the shotD

9.3 Continuity over years and locations

&ater scenes follo# 2ateo through different periods of life and across Guate%ala and 1#eden" The challenge is not to force identical pi*els across decades, but to preserve identit(cues and narrative continuit(#hile allo#ing age, costu%e and conte*t to change deliberatel(" The database therefore needs both invariants and versioned character states"

9.4 Technical environments and controlled complexity

The pro'ect also includes scientific and technological environ%ents, including a biological co%puting concept" These scenes test #hether the s(ste% can preserve designed %achiner(and spatial relationships #hile directing local reactions or state changes" Again, the re<uire%ent is selective transfor%ation, not auto%atic redesign"

9.5 A production consequence

The i%%ediate production strateg(is therefore prag%atic" The fil% can first e*ist as a carefull(ti%ed se<uence of validated i%ages, dialogue, sound, cuts and selected ca%era effects" 2otion is added onl(#here it contributes" This creates a coherent fil% structure before as$ing generative s(ste%s to ani%ate ever(fra%e"

10. Openness, rights, provenance and human direction

10.1 Open does not mean model-free

An open fil%%a$ing environ%ent %a(use proprietar(or hosted %odels #hen useful" Openness here refers pri%aril(to the durabilit(and intelligibilit(of the pro'ect: standard %edia files, e*portable %etadata, e*plicit rights and provenance, replaceable adapters, and the abilit(to continue editing if a provider disappears"

10.2 Provenance should be part of production, not an afterthought

As s(nthetic %edia beco%es nor%al, a fil% pro'ect should be able to record #hich assets #ere hu%an!created, AI! assisted, generated, licensed, transfor%ed or co%posited@ #hich %odel or service produced a render@ and #hich hu%an decisions governed the output" Provenance is therefore closel(related to the sa%e database architecture needed for continuit("

10.3 Human direction should be inspectable

A hu%an!in!the!loop clai% is #ea$ if the hu%an contribution e*ists onl(as a vague pro%pt" 0IRECTOR and Cine&ingo ai% to %a$e direction inspectable: selected i%ages, e*plicit %ass, locs, ti%ing, behaviours, re'ected renders and approved states" This is both a creative and governance advantage"

10.4 European context

European transparenc(re<uire%ents for certain AI!generated or %anipulated content beca%e applicable in August ,;,6" The e*act legal treat%ent depends on conte*t and i%ple%entation, but the broader production lesson is alread(clear: provenance, disclosure and docu%entation are beco%ing part of audiovisual #or$flo# design rather than optional %etadata added at the end"

11. Technology watch: October 2026

The %ost relevant signals for CineAI &ab are not si%pl(i%prove%ents in spectacle" The(are signs that the ecos(ste% is %oving to#ard stronger anchoring, faster iteration, integrated editing and engine abstraction"

Date	Signal	Why it matters for CineAI Lab
34 Oct ,;,6	Co%f(.I KCI0IA C9+ fra%e!generation nodes interpolate bet#een ad'acent fra%es"	E*isting fra%es are treated as anchors, #hich is conceptuall(closer to the flipboo$ %odel than unconstrained #hole!shot invention"
, Oct ,;,6	Eu%an Provenance in 9il% consultation update tests production declarations"	Provenance can be designed into a pro'ect database rather than handled onl(at distribution"
4; 1ep ,;,6	Co%f(API pac$ages #or$flo#s as deplo(able services"	The e*ecution la(er can %ove behind a fil%%a$er!facing interface"
,8 1ep ,;,6	/an ,"7 docu%entation lists first!fra%e, first!and! last!fra%e, and continuation %odes"	True first last!fra%e anchoring %oves closer to the desired $e(fra%e #or$flo#"
,4 1ep ,;,6	Co%f(Router e*poses %ultiple %edia %odels through one API"	2odel abstraction supports engine independence"
3; 1ep ,;,6	Run#a(research targets lo#er ti%e!to!first!fra%e and strea%ing generation"	9ast feedbac$ is essential for repeated s%all directorial corrections"

8 1ep , ; ,6	Adobe 9irefl(beta integrates generation, ele%ent! level editing, assets and video asse%bl("	AI is %oving fro% isolated generators to#ard structured creative #or$spaces"
3, Aug , ; ,6	&T+!,"- e*pands open local video #or$flo#s"	&ocal e*ecution re%ains strategicall(i%portant even #hen current <ualit(or speed is insufficient"
43 Ful , ; ,6	1eedance ,"- e%phasises reference!based %ulti%odal generation and longer se<uences"	Reference control and se<uence!level thin$ing are beco%ing %ainstrea% %odel ob'ectives"
, Aug , ; ,6	E . AI transparenc(obligations beco%e applicable"	1(nthetic!%edia provenance increasingl(intersects #ith production architecture"

11.1 A field moving toward the missing layer

These develop%ents do not prove that the 0IRECTOR proble% is solved" The(do suggest convergence" 9irst last!fra%e control, interpolation, strea%ing generation, unified #or$spaces and %ulti!%odel APIs all reduce the distance bet#een toda()s generators and a true fil%%a$ing environ%ent" The re%aining <uestion is #ho o#ns the pro'ect %e%or(and directorial logic: the %odel interface, or the fil%%a$er"

11.2 Selected current sources

Co%f(. I, KCI0IA C9+ nodes: https: blog"co%f("org p nvidia!vf*!nodes!for!video!upscaling

Eu%an Provenance in 9il%: https: hu%anprovenance"fil% ne#s consultation!update!septe%ber

Co%f(. I blog archive API announce%ents: https: blog"co%f("org archive

Alibaba Cloud, / an i%age!to!video guide: https: ###"alibabacloud"co% help en %odel!studio #an!i%age!to!video! guide

Co%f(Router: https: blog"co%f("org p introducing!co%f(!router!one!api

Run#a(, To#ards instant video generation: https: run#a("co% ne#s research to#ards!instant!video!generation

Adobe 9irefl(unified generation and editing: https: help*"adobe"co% fr firefl(#eb unified!generation!and!editing! e*perience generation!and!editing!e*perience!overvie#"ht%l

Co%f(. I &T+!,"- support: https: blog"co%f("org p lt*!,-!da(!:!support!in!co%f(ui

B(te0ance 1eedance ,"-: https: seed"b(tedance"co% en blog one!ta$e!creation!fle*ible!referencing!introducing! seedance!,!-

European Co%%ission, AI Act transparenc(guidelines: https: digital!strateg("ec"europa"eu en librar(guidelines! transparenc(!obligations!providers!and!deplo(ers!ai!s(ste%s

12. Evaluation framework for controllable AI filmmaking

12.1 Quality should be measured against directorial intent

A beautiful output can still be a failure if it changes the #rong thing" Evaluation should therefore co%pare the render against e*plicit shot constraints, not onl(against generic visual <ualit("

Dimension	Question	Example acceptance criterion
Identit(	0id the intended character re%ain recognisable and consistentD	Ko unintended face drift@ age changes onl(#hen specified"
Co%position	0id the ca%era and spatial arrange%ent re%ain as directedD	&oc$ed ca%era and fra%ing re%ain unchanged"
Ob'ect per%anence	0id props and designed ob'ects surviveD	PuBBle piece, table or %achiner(re%ains present and stable unless intentionall(%oved"
1elective %otion	0id onl(authorised regions %oveD	Eand gesture occurs #ithout unrelated bac$ground or costu%e change"

Te%poral accurac(	0id actions occur at the intended ti%e and durationD	GaBe begins #ithin the specified cue #indo#"
Perfor%ance	0oes %otion read as intentional acting rather than noiseD	Pause, gaBe and gesture support the dra%atic beat"
Continuit(	0oes the shot re%ain co%patible #ith ad'acent shotsD	Costu%e, lighting and ob'ect state %atch the se<uence"
Iteration cost	Can the fil%%a$er revise the shot at practical speedD	1%all changes do not re<uire rebuilding unrelated content"
Provenance	Can the origin and transfor%ation histor(of assets be reconstructedD	Render lin$s bac$ to source assets, instructions and engine 'ob"
Portabilit(	Can the pro'ect survive a change of engineD	Core shot representation does not depend on one %odel graph"

12.2 Failure should be informative

A useful production s(ste% should not onl(sa(that a render failed" It should identif(#hat constraint #as violated: identit(drift, ca%era %ove%ent, hand anato%(, bac$ground defor%ation, te%poral %is%atch or another categor(" This transfor%s re'ected generations into structured evidence for both creative iteration and %odel evaluation"

13. Roadmap

13.1 Near-term: finish the film structure before perfecting animation

- Consolidate The 2issing Piece as a co%plete i%age!first se<uence #ith validated i%ages, durations, dialogue and sound"
- Ieep the canonical one!i%age!per!shot idea discipline and 36:: fra%ing for fil% assets"
- . se dedicated close!up i%ages rather than cropping or reusing #ide shots #hen the dra%atic function changes"
- Ani%ate selectivel(, beginning #ith shots #here %ove%ent clearl(adds infor%ation or e%otion"

13.2 DIRECTOR prototype

- Co%plete the persistent asset and shot!instance %odel"
- Ieep selections na%ed, %ovable, re%ovable and stored independentl(fro% source i%ages"
- E*pose %as$s, feathering, loc$s and ani%atable regions in the Inspector"
- Represent dialogue, instructions, errors to avoid and ti%eline events as separate editable la(ers"
- Allo# engine adapters to consu%e the sa%e shot representation"

13.3 CineLingo development

- Parse natural!language direction into e*plicit ob'ects, ti%e references and actions"
- Calidate contradictions before rendering"
- 2aintain a reusable behaviour librar(for co%%on cine%atic perfor%ance patterns"
- Co%pile the sa%e behaviour into different engine!specific i%ple%entations"
- Preserve the fil%%a$er!facing description even #hen an engine changes"

13.4 Engine research

- Test true first last!fra%e #or$flo#s as the(beco%e practicall(accessible"
- Co%pare interpolation approaches #ith generative transition approaches for local %icro!%ove%ent"
- 2easure ti%e!to!first!previe# and revision latenc(, not onl(final visual <ualit("
- Prefer engines that e*pose region control, $e(fra%es, ca%era loc$s and deter%inistic constraints"
- Avoid rebuilding the CineAI architecture around a single rapidl(changing %odel fa%il("

13.5 Research questions

- Eo# %uch of a shot can re%ain literall(unchanged #hile selected regions are generatedD
- Can first last!fra%e constraints preserve identit(better than start!i%age!onl(conditioningD
- / hat representation of gaBe, gesture and e*pression is portable across video %odelsD
- Eo# should an authoring tool distinguish hard loc$s fro% soft preferencesD
- Can partial re!rendering reduce both generation cost and continuit(errorsD
- / hich provenance data should be stored b(default for an AI!assisted fil%D
- / hat is the %ini%u% interface a fil%%a$er needs to direct AI #ithout beco%ing a node!graph engineerD

Conclusion

CineAI &ab began #ith the e*cite%ent and frustration of e%erging AI %edia tools" The e*peri%ents confir%ed that i%ages, speech and %otion can increasingl(be generated" The(also %ade a different proble% i%possible to ignore: fil%%a$ing depends on continuit(of intention"

The central proposal of version ;"3; is therefore not a ne# generative %odel" It is an authoring architecture" Approved i%ages beco%e persistent assets" Their uses beco%e shot instances" 1elections, %as$s and loc$s define #hat %a(change" Cine&ingo describes events over ti%e" AI engines e*ecute behind the interface and can be replaced" The fil% survives the%"

This is a %odest clai% co%pared #ith pro%ises of auto%atic cine%a, but it %a(be a %ore durable one" The goal is not to as$ artificial intelligence to beco%e the director" The goal is to give the director a better instru%ent"

> **CineAI Lab, October 2026**
>
> The fil% is the pro'ect" The interface re%e%bers the direction" The engine perfor%s a tas$"

Appendix A. Director/Lingo to CineAI mapping

Historical concept	CineAI interpretation	Reason retained
Cast	2 edia Assets	1eparates source resources fro% their uses in shots"
1tage	/ =1l / =G 1tage	2 a$es the intended co%position e*plicit and editable"
1core	Ti%eline 1e<uence	Represents #hen instances and events e*ist"
Cast %e%ber	1ource %edia	Preserves originals independentl(fro% shot place%ent"
1prite	1hot instance	Gives each placed use its o#n transfor%, ti%ing and constraints"
Behavior	Cine%atic behaviour	Encodes reusable acting or %otion intentions"
9ra%e script	Te%poral event	Attaches actions to %o%ents or intervals"
1prite script	Instance behaviour	Targets one placed character or ob'ect"
2 ovie script	Pro'ect behaviour	Represents pro'ect!#ide rules or policies"
&ingo	Cine&ingo	Adds structured behaviour #hen visual la(out alone is insufficient"
Pla(bac$ head	Pla(head	I eeps ti%e as a native part of direction"
Propert(list	1tructured para%eters	1upports engine!neutral representation of constraints"

Appendix B. Minimal data model

This sche%a is intentionall(conceptual" I%ple%entation details %a(evolve, but the separation bet#een durable fil% data and replaceable engine e*ecution should re%ain"

Appendix C. Experiment matrix

Experiment	What worked	Observed limit	Architectural lesson

Calidated stills	1trong control of i%age identit(, co%position and production design	Ko inherent %otion	Treat approved i%ages as persistent assets"
,"-0 ca%era paralla*	0eter%inistic ca%era and co%positing	Read as a %oving still #hen perfor%ance #as needed	1eparate ca%era %otion fro% actor ob'ect behaviour"
1(nthetic voice	1peech generation and ti%ing	Perfor%ance nuance re%ained generic in difficult scenes	0ialogue and acting direction need separate la(ers"
Cise%es lip!s(nc	2 outh!shape ti%ing can be structured	9acial anato%(and identit(instabilit(	. se region!specific control and perfor%ance constraints"
&T+!,"- local	Open local inference technicall(possible	2 e%or(pressure, blur ca%era instabilit(, iteration speed	&ocal e*ecution %atters, but engine %ust re%ain replaceable"
Co%f(. I / an ,", tested #or$flo#	Real I,C clips fro% a source i%age	1tart!i%age!onl(control, #hole!shot drift, long iterations	Keed end!i%age anchors, loc$s and selected %otion"
0IRECTOR Cine&ingo	9il% can be represented as assets L instances L behaviours	Interface and adapters still under develop%ent	2 ove the creative source of truth above the engine la(er"

Appendix D. Selected references

CineAI Lab public site: https: cineai!lab"org

CineAI Lab GitHub repository: https: github"co% ReneBoroughs cineai!lab

Adobe Director Support Center archive: https: ###"adobe"co% support director do#nloads"ht%l

Director Scripting Reference, Macromedia Director MX 2004: https: eclass"h%u"gr %odules docu%ent file"php TP3:5 dr%*,;;5HscriptingHref"pdf

Cardiff University, Lingo / Director basics: https: users"cs"cf"ac"u$ 0ave" 2 arshall 2 ulti%edia P09 &ingoHBasics"pdf

ComfyUI blog: https: blog"co%f("org

Alibaba Cloud Wan image-to-video documentation: https: ###"alibabacloud"co% help en %odel!studio #an!i%age! to!video!guide

Runway Research: https: run#a("co% research

Adobe Firefly: https: ###"adobe"co% products firefl("ht%l

European Commission AI Act transparency guidance: https: digital!strateg("ec"europa"eu en librar(guidelines! transparenc(!obligations!providers!and!deplo(ers!ai!s(ste%s

Human Provenance in Film: https: hu%anprovenance"fil%

Project documents informing this edition

- CIKEAIH0IRECTORH&IKGOH / =1I / =GHRE9EREKCE"%d M #or$ing architecture reference, 36 August ,;,6"
- CIKE&IKGOH1PECHC;3"%d M initial functional specification, 3 1epte%ber ,;,6"
- CineAIH&abH&T+,-HAppleH1iliconHTechnicalHReport"doc* M local inference e*peri%ent, 3- August ,;,6"
- The 2 issing Piece stor(boards and scene docu%ents M October ,;,6 #or$ing %aterials"
- CineAI &ab #ebsite research, e*peri%ent and #atch pages M updated 38 October ,;,6"

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